Limits to Arbitrage in the Indian Equity Market

First Results

Prince Kumar (23B3032), BS Economics, IIT Bombay
Guide: Prof. Aditi Chaubal | EC 495 / EC 497
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1 Summary

Since the October report I have done four things.

1. **Fixed the test plan in writing and froze it** before relating any mispricing measure to bans, reforms or any other treatment. Where the plan left a detail open, I wrote the detail down with a date before running the test that needed it. Results that I added after seeing a test are reported as exploratory.
2. **Ran every test in the plan**: four main hypotheses, corrected jointly for multiple testing, and ten secondary ones.
3. **Estimated the no-arbitrage band** with three-regime threshold and smooth-transition models, over time and across states, and compared it with a bottom-up cost band.
4. **Built a constraint index** for every NSE stock and quarter from frictions alone, and checked it against price delay and earnings drift.

The main findings: bans bind hard on trading; correction of the basis does not become one-sided in the way the model predicts; the basis falls when a ban starts; the band is wide and asymmetric, and its edges do not move one-for-one with the tax; entry into the F&O segment speeds up the response of prices to market news.

2 Two corrections to the October report

- *Stock lending.* The October report said that only F&O stocks are eligible for short selling and stock lending. Short-selling eligibility is indeed limited to F&O stocks (SEBI circular of 5 January 2024), but the securities lending and borrowing segment has also covered other liquid stocks since July 2013. In the daily SLB files, non-F&O stocks account for 6% to 23% of lending trades in each year from 2019. SLB trading remains thin, so the practical overnight short is still the stock future.
- *Dates of results.* I now date quarterly results from the exchange's first dissemination time, combining its announcements, the results filings and board-meeting dates, and map each result to the symbol the stock traded under on that day. The filings alone often carried later re-filing times. With this and a few other checks, some counts have changed slightly: 1,788 ban spells and 9,416 ban stock-days (October report: 1,789 and 9,414), and 8,594 results announcements for F&O stocks (7,910).

3 Data added

Daily securities-lending files (2015 to 2026), adjusted daily returns and market capitalisations for all NSE stocks, and AMFI's monthly flows and assets of arbitrage funds (April 2018 to September 2026). The basis sample has 486,040 stock-days for 371 stocks, 8,851 of them in a ban.

4 Main hypotheses

Table 1: The four main hypotheses (Holm correction at 5%)

Hypothesis	Main estimate	Result
Bans move trading out of futures	Futures contracts −2.00 log points (s.e. 0.11); delivery share +0.041	Supported
Costs set the band edges	80th percentile +6.4 and +7.4 bp around the 2024 and 2026 STT increases	Supported on the planned statistic; weak against placebo dates
Short-sale access (F&O entry and exit)	Price delay after entry −0.109 (s.e. 0.023)	Supported through price delay; drift after bad news goes the other way
One-sided correction in bans	Change in reversion in a ban: upper regime −0.089, lower +0.081	Rejected, opposite sign

Trading in a ban. In a ban, futures trading falls by about 87%, cash volume rises by about 7.5% and more of it is delivered. The model also predicts that every futures trade in a ban closes a long and a short, so the change in open interest equals minus the volume. This holds almost exactly from 23 March 2020, when SEBI's measures made the clearing corporations judge ban violations on peak intraday positions, and not before, when only end-of-day positions counted (Figure 1).

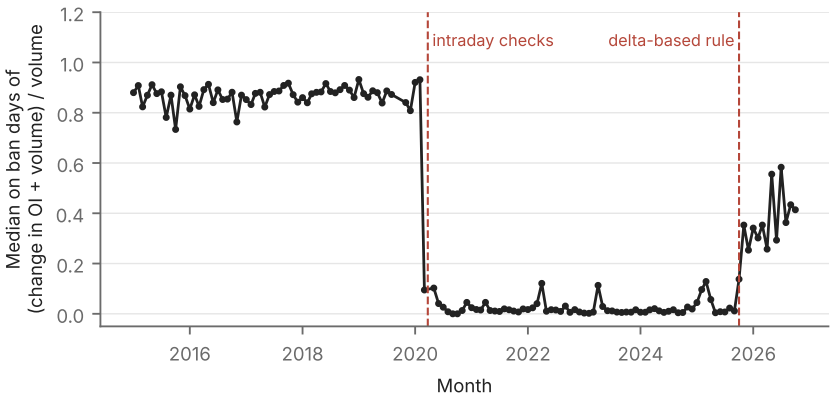


Figure 1: Monthly median of (change in open interest + volume) / volume on ban days. A value of zero means every trade closed positions on both sides.

Correction in a ban. The model predicts that rich futures stop being corrected in a ban, because nobody may open new cash-and-carry positions. In the data, rich futures revert faster

in a ban and cheap futures more slowly. The pattern is concentrated in the first days of a spell and survives a correction for noise in daily prices. A reading consistent with it, though formed after the test, is that at the start of a ban the model's second channel dominates: demand for new futures exposure is blocked, which removes the pressure that had made futures rich.

Costs. Around the 2024 and 2026 STT increases the upper part of the basis distribution rose, and the ordering of the three increases matches the size of the cost changes. At 30 placebo dates, however, the same statistic varies by about 7 bp, because shocks common to all stocks move the whole distribution; the rise is mostly such a shift, and larger than the cost changes.

F&O entry and exit. After a stock enters the F&O segment its price responds much faster to market-wide news, and after an exit somewhat more slowly. Drift after bad news becomes stronger, not weaker, after entry. Entry requires a stock to have grown, which by itself speeds price adjustment; the leads before entry are flat, but this cannot be ruled out.

5 Secondary results

The level of the basis when a ban starts. Against stocks that came close to 95% but did not cross it, matched on their utilisation path and basis, the basis falls by about 6 bp in the first days of a ban and recovers after about a week (Figure 2). The fall is larger when open interest had been building fast before entry, as the model's level result implies; the planned split (rising against falling open interest) has the predicted signs but only 30 events with falling open interest ($p = 0.08$), while a median split and a continuous measure are significant.

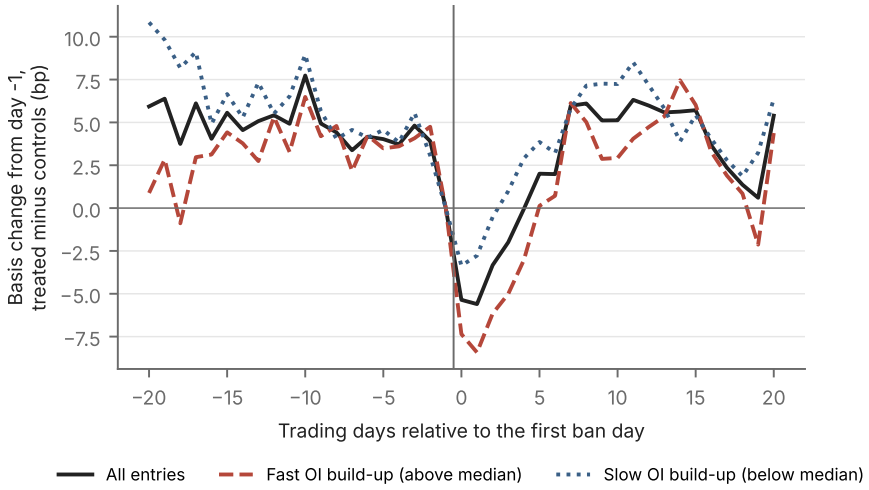


Figure 2: Change in the basis from the day before a ban starts, banned stocks minus matched controls.

The 95% trigger. Utilisation does not bunch below the trigger (density test $p = 0.86$); it accelerates toward it. At the trigger, futures trading drops sharply, while the basis is already lower on the day utilisation crosses 95%. As utilisation approaches the trigger on days without a ban, the basis is more often rich (Figure 3). Between 80% and 95%, at equal utilisation and recent basis, being in a ban does not move the basis further.

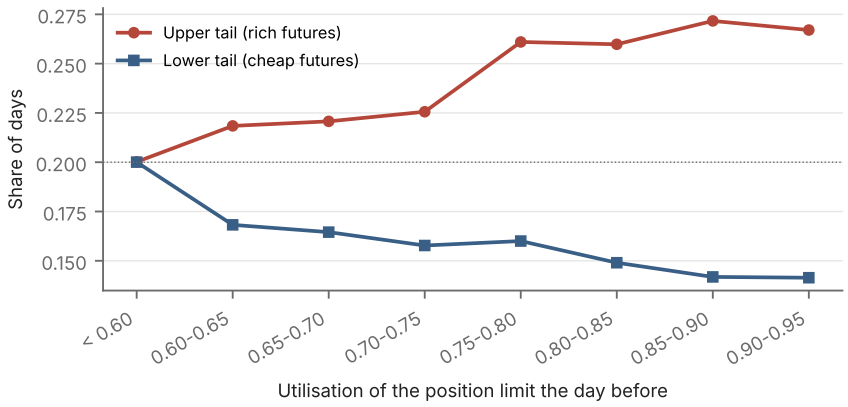


Figure 3: Share of days with the basis in the stock's upper and lower tail, by utilisation the day before (days without a ban, before October 2025; raw shares). With date effects removed the upper-tail rise remains and the lower-tail decline disappears.

Other secondary tests. The October 2025 change in the position-limit formula moved utilisation strongly; its effect on the basis is in the predicted direction but not significant. Bad news in bans could not be tested with useful precision (55 events). Across stocks, higher frictions go with wider bands. Inflows into arbitrage funds are followed by a lower upper tail of the basis, but the evidence is weak. Anomaly returns do not show the predicted asymmetry between constrained and unconstrained stocks once returns are value-weighted.

6 The no-arbitrage band

I estimate three-regime threshold models of the daily change in the basis on the lagged deviation from each stock's median (non-ban days, 473,110 observations), with a bootstrap test of linearity and bootstrap intervals for the thresholds, and compare them with smooth-transition models.

- The band is about -35 to $+24$ bp around each stock's median, and linearity is clearly rejected. The threshold model fits best; a logistic smooth-transition model, which allows different speeds below and above, fits almost as well; a symmetric exponential one fits worse.
- Noise in daily prices inflates the measured speed of reversion. Instrumenting the lagged deviation with its own lag removes most of the apparent reversion inside the band and most of the difference between the two outer regimes.
- The band is wider far from expiry; reversion is faster in the last week before expiry, slower when utilisation is high, and faster when arbitrage-fund assets are large (in smooth-transition models with these states).
- A new cash-and-carry round trip costs 27 to 31 bp (STT, stamp duty, exchange charges and brokerage), which places it in the outer tail of the basis, while rolling an existing position costs 4 to 8 bp. This is one reason why estimated edges sit inside the full cost band.

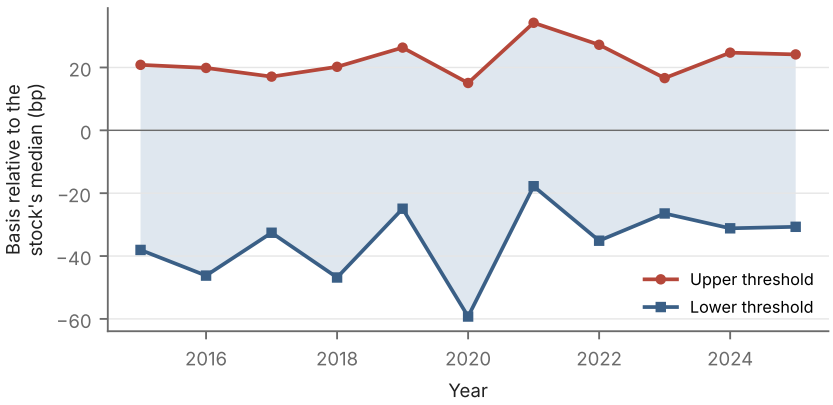


Figure 4: Estimated band edges by year (basis relative to each stock's median).

7 A constraint index

For all 3,310 NSE stocks and each quarter from 2015 to 2026, the index combines five frictions: no short channel, scarcity of stock lending, the Corwin-Schultz spread, Amihud illiquidity and idiosyncratic volatility (first principal component, 53% of their variance, stable loadings). More constrained stocks respond more slowly to market news the next year, also within a stock over time, and drift more after bad but not good news. Within F&O stocks a separate index that adds utilisation of the position limit is the relevant measure for the futures-cash band.

8 Plan from here

1. Complete the thesis chapters; a full draft with all results exists, and I can share it.
2. Read the three papers I could not obtain online (Saurav, Srivastava and Agarwalla 2026; Gagnon 2018; Brennan and Schwartz 1990) and position the ban results against the first.
3. Revisit the 2025 reform as more post-reform data arrive.

I would value your views on two points: whether the interpretation of the one-sided-correction result (blocked new demand at the start of a ban) should be developed into an extension of the model, and whether the smooth-transition models with observed states should be taken further for the band chapter.